M201103W- Signals and Systems

ssch7_3 –Properties of unilateral z-transform

14:10 pm Wednesday @ C107, 1st Teaching Building

14:10 pm Friday @ A107, 1st Teaching Building

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Properties of unilateral *z*-transform

- Linearity
- Time Shifting
- Convolution
- Summation
- Exponential weighting
- Linear weighting

(Differentiation in the z-domain)

- Reversal in time domain
- Initial- and final- value theorems

Properties of unilateral *z*-transform

- Linearity

$$x[k] \xleftrightarrow{\mathcal{Z}} X(z), \qquad |z| > R_x$$

$$x_1[k] \xleftrightarrow{\mathcal{Z}} X_1(z), \qquad |z| > R_{x1}$$

$$x_2[k] \xleftrightarrow{\mathcal{Z}} X_2(z), \qquad |z| > R_{x2}$$

$$ax_1[k] + bx_2[k] \xleftrightarrow{\mathcal{Z}} aX_1(z) + bX_2(z),$$

$$|z| > \max(R_{x1}, R_{x2})$$

Properties of unilateral *z*-transform

►Time Shifting

If $$x[k]u[k] \xleftrightarrow{\mathscr{Z}} X(z), |z| > R_x$$

※ For causal signals,

$$x[k-n]u[k-n] \xleftrightarrow{\mathscr{Z}} z^{-n}X(z), \quad |z| > R_x$$

Time shifting by n leads to multiplication by z^{-n} in z-Transform

※ For incausal signals,

$$\mathscr{Z}\{x[k+n]\,u[k]\} = z^{n}\left[X(z) - \sum_{k=0}^{n-1} x[k]z^{-k}\right], \quad |z| > R_x$$

$$\mathscr{Z}\{x[k-n]\,u[k]\} = z^{-n}\left[X(z) + \sum_{k=-n}^{-1} x[k]z^{-k}\right], \quad |z| > R_x$$

Time Shifting

Right shifting

$$x[k-n]u[k] \longleftrightarrow z^{-n}[X(z)+\sum_{k=-n}^{-1} x[k]z^{-k}]$$

$$\mathcal{Z}\{x[k-1]u[k]\}=z^{-1}X(z)+x[-1]$$

$$\mathcal{Z}\{x[k-2]u[k]\}=z^{-2}X(z)+z^{-1}x[-1]+x[-2]$$

Left shifting

$$x[k+n]u[k] \longleftrightarrow z^{n}[X(z)-\sum_{k=0}^{n-1} x[k]z^{-k}]$$

$$\mathcal{Z}\{x[k+1]u[k]\} = zX(z)-zx[0]$$

$$\mathcal{Z}\{x[k+2]u[k]\} = z^{2}X(z)-z^{2}x[0]-z^{1}x[1]$$

Example. Calculate z-transform of $x[k]=u[k]-u[k-N]$. Determine the ROC.

Calculation according to definition, or time shifting property

Solution: $$u[k] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-z^{-1}}, \qquad |z|>1$$

$$u[k-N] \xleftrightarrow{\mathcal{Z}} \frac{z^{-N}}{1-z^{-1}}, \qquad |z|>1$$

$$X(z)=\frac{1}{1-z^{-1}}-\frac{z^{-N}}{1-z^{-1}} = \frac{1-z^{-N}}{1-z^{-1}}$$

ROC is $|z| > 0$ because $x[k]$ is an finite-length signal.

Enlargement in ROC

Example. The definition of autocorrelation of $x[n]$ is

$$R_x[n] = \sum_{k=-\infty}^{+\infty} x[k]x[n+k] \text{ . Calculate } \mathscr{Z}\{R_x[n]\}$$

$$\mathscr{Z}\{R_x[n]\} = \sum_{k=-\infty}^{+\infty} x[k]\mathscr{Z}\{x[n+k]\}$$

$$= \sum_{k=-\infty}^{+\infty} x[k]X(z)z^k = X(z)\sum_{k=-\infty}^{+\infty} x[k]z^k$$

$$= X(z)X(z^{-1})$$

$$R_x[n] = x[n]*x[-n] \qquad \mathscr{Z}\{R_x[n]\} = X(z)X(z^{-1})$$

Properties of unilateral *z*-transform

- Convolution property

$$x_1[k]u[k] * x_2[k]u[k] \xleftrightarrow{\mathcal{Z}} X_1(z)\, X_2(z) \qquad |z| > \max(R_{x1}, R_{x2})$$

$$\mathcal{Z}\left\{x_1[k]u[k] * x_2[k]u[k]\right\} = \mathcal{Z}\left\{\sum_{n=-\infty}^{+\infty} x_1[n]u[n] \cdot x_2[k-n]u[k-n]\right\}$$

$$= \sum_{n=0}^{+\infty} x_1[n]\mathcal{Z}\left\{x_2[k-n]u[k-n]\right\} = X_2(z)\sum_{n=0}^{+\infty} x_1[n]z^{-n}$$

$$= X_1(z)X_2(z), \quad |z| > \max(R_{x1}, R_{x2})$$

z- Transform of convolution of two D-T signals equals multiplication of z-Transform of original signal respectively.

Example. Calculate $x[k] = a^k u[k] * b^k u[k] \quad a \neq b$ using the convolution property of z-Transform.

Solution: According to convolution property of z-Transform.

$$\mathcal{Z}\left\{x[k]\right\} = \mathcal{Z}\left\{a^k u[k] * b^k u[k]\right\} = \mathcal{Z}\left\{a^k u[k]\right\} \cdot \mathcal{Z}\left\{b^k u[k]\right\}$$

Because $\mathcal{Z}\left\{a^k u[k]\right\} = \dfrac{z}{z-a}, \ |z| > |a| \qquad \mathcal{Z}\left\{b^k u[k]\right\} = \dfrac{z}{z-b}, \qquad |z| > |b|$

Thus $X(z) = \mathcal{Z}\left\{x[k]\right\} = \dfrac{z^2}{(z-a)(z-b)} \qquad |z| > \max\{|a|, |b|\}$

$$\frac{X(z)}{z} = \frac{z}{(z-a)(z-b)} = \frac{A}{z-a} + \frac{B}{z-b} \qquad X(z) = \frac{Az}{z-a} + \frac{Bz}{z-b},$$

$$x[k] = Aa^k u[k] + Bb^k u[k] = \frac{1}{a-b}(a^{k+1} - b^{k+1})u[k] \qquad a \neq b$$

Example. If $x[k]$ is causal, calculate z-transform of $\sum_{n=0}^{k} x[n]$

Solution $$\sum_{n=0}^{k} x[n] = x[k] * u[k]$$

If $$x[k] \xleftrightarrow{\mathcal{Z}} X(z), \qquad |z| > R_x$$

Using convolution property of z-Transform and:

$$u[k] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-z^{-1}}, \qquad |z| > 1$$

Therefore:

$$\mathcal{Z}\left\{\sum_{n=0}^{k} x[n]\right\} = \mathcal{Z}\left\{x[k]\right\} \cdot \mathcal{Z}\left\{u[k]\right\} = \frac{X(z)}{1-z^{-1}}, \qquad |z| > \max(1, R_x)$$

Properties of unilateral *z*-transform

- Summation property

If $$x[k]u[k] \xleftrightarrow{\mathscr{Z}} X(z), \qquad |z| > R_x$$

Then $$\sum_{n=0}^{k} x[n] \xleftrightarrow{\mathscr{Z}} \frac{1}{1-z^{-1}} X(z), \qquad |z| > \max(R_x, 1)$$